

KUNAL CHANDRA

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SUMMARY

Software Development Engineer candidate with expertise in programming, data structures, algorithms, and AI/ML, delivering measurable, high-impact solutions through rigorous problem-solving.

EXPERIENCE

Defence Research and Development Organisation (DRDO) Jun 2026 – Present

Software Development Engineer Intern, *Institute of Nuclear Medicine and Allied Sciences (INMAS)* Delhi

- Engineered an **OCR-based visitor gate pass system** for real-time ID verification and automated generation.
- Accelerated document processing pipelines using **GPU-based OCR** and advanced image preprocessing techniques.
- Developed an **LLM-powered RAG chatbot** to streamline intelligent internal document retrieval.

Defence Research and Development Organisation (DRDO) Jan 2026

Machine Learning Intern, *Solid State Physics Laboratory (SSPL)* Delhi — [Certificate](#)

- Designed a **Siamese Nested U-Net (SNUNet)** with ECAM in **PyTorch** for satellite change detection.
- Constructed a 6-class **semantic change classification** pipeline for defense surveillance and infrastructure monitoring.
- Attained **96.37% Precision, 95.95% Recall, and 96.16% F1-Score** on the **CDD dataset**.

Intelligence Fusion & Strategic Operations (IFSO) Special Cell, Delhi Police Aug 2025

Software Development Engineer Intern Delhi — [Certificate](#)

- Developed a modular **Social Media Chat Parser and Analyzer** for WhatsApp and Instagram forensic data extraction.
- Implemented a decryption pipeline using **Pyzipper** and **Wadecrypt** to process encrypted WhatsApp backups.
- Devised a heuristic parsing engine with **BeautifulSoup** to consolidate malformed Instagram data.

PROJECTS

SMARTFLOW: Intelligent Traffic Signal Control System | YOLOv8, BYTETrack, ESP32, RFID [Live](#)

- Engineered an **AI-driven traffic signal system** using **YOLOv8** vehicle detection and **BYTETracker** multi-object tracking.
- Implemented **virtual line-based density estimation** with dynamic annotations for adaptive signal timing.
- Integrated **ESP32 and RFID** modules to enable emergency and priority vehicle detection at intersections.

Pravha: AI-Powered Flood Forecasting & Emergency Response | FastAPI, React, MongoDB, Docker, XGBoost [Live](#)

- Developed an **AI-powered flood forecasting platform** using **XGBoost**, achieving **86.5% prediction accuracy**.
- Delivered multilingual flood alerts in **12 Indian languages** with **GIS-based flood-zone** and shelter routing.
- Engineered a **Mesh-network based SOS** app using **Bluetooth Low Energy (BLE)** for peer-to-peer emergency alerts.

Enterprise Knowledge Assistant using RAG and Local LLMs | Llama.cpp, ChromaDB, LangChain, FastAPI, React

- Architected an **incremental indexing pipeline** with document-level version hashing to re-embed only modified files, cutting redundant compute.
- Implemented **query rewriting** and **tree-summarization** synthesis with **Llama.cpp** and **ChromaDB** for accurate retrieval.
- Built a **FastAPI and React (TypeScript)** interface with persistent memory for context-aware, multi-turn Q&A.

EDUCATION

Haridwar University Sep 2023 – Jun 2027

Bachelor of Technology in Computer Science and Engineering CGPA: 8.00

Indian Institute of Technology Mandi June 2025 – Oct 2026

Minor in Artificial Intelligence and Data Science CGPA: 8.70

Sam International School Class X: 94% — Class XII: 80%

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C, SQL

Frameworks & Libraries: Pandas, NumPy, Matplotlib, Seaborn, NLTK, FastAPI, Flask, PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, MediaPipe, REST APIs, JWT Authentication, API Rate Limiting

Generative AI & LLMs: LangChain, LangGraph, LlamaIndex, Hugging Face Transformers, Sentence Transformers, RAG, ChromaDB

Databases & Developer Tools: PostgreSQL, MongoDB, MySQL, Git, GitHub, Docker, Kubernetes, CI/CD (GitHub Actions)

ACHIEVEMENTS & CERTIFICATIONS

- Cleared CodeVita Round 2 with rank **270 among 100,000+** participants [Link](#)
- Reached **Top 25 Finalists** at Trwizardthon Hackathon among **1,300+ teams, 5,200+ students, 28+ states** [Link](#)
- Achieved **Top 10 out of 300+ teams** at AI Manthan 2025 **National Hackathon** hosted by IIT Delhi [Link](#)
- Earned Oracle Cloud Infrastructure 2025 Certified **DevOps** [Link](#) and **Data Science** [Link](#) Professional certifications